MDM

Subject :- ME24052 – Introduction to Robotics & Automation

Topic:- End Effectors

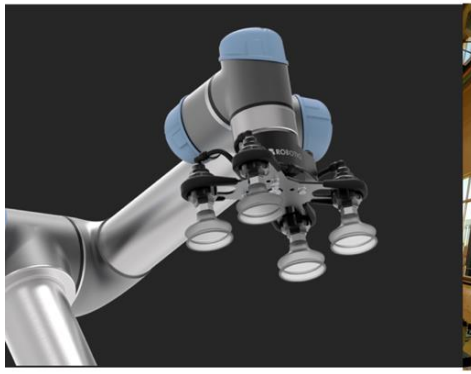


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Unit-3: End Effectors	**[07 Hrs.]**
Grippers, Mechanical Grippers, Pneumatic and Hydraulic- Grippers, Magnetic Grippers, Vacuum Grippers; Two Fingered and Three Fingered Grippers; Internal Grippers and External Grippers; Advance Grippers- Adaptive grippers, Soft Robotics Grippers, Tactile Sensor Grippers; Various process tools as end effectors; Robot end effectors interface, Active and passive compliance, Selection and Design Considerations.	

Grippers

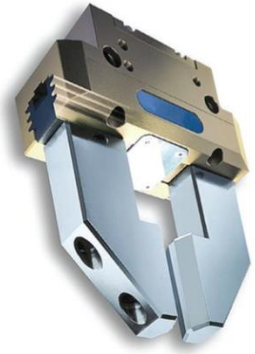
- Grippers are a type of **end effector** used in robotics to interact with objects by gripping, holding, or manipulating them.
- The design and selection of a gripper depend on the task at hand, the object's size, shape, material, and weight.
- Various types of grippers, from mechanical to advanced soft robotic grippers, are employed for different applications.



Vacuum gripper



Magnetic gripper



Hydraulic gripper



Electric gripper

Mechanical Grippers

Principle

- Mechanical grippers use a mechanical mechanism, typically involving gears, cams, or levers, to grip an object. They convert the actuation force (from a motor, pneumatic, or hydraulic source) into gripping force, enabling the end effector to grasp the object securelv.



Types of Mechanical Grippers

- **Two-Finger Grippers**: Most common type; involves two opposing fingers that can move toward or away from each other to grip an object.
- **Applications:** Used for handling regular-shaped objects such as cylinders, boxes, or simple geometries.



- **Three-Finger Grippers**: Offers more stable gripping for irregularly shaped objects; the fingers move symmetrically to grip from three sides.
- **Applications:** Used in robotics and automation where irregular or fragile items need to be handled.



Pneumatic and Hydraulic Grippers

Pneumatic Grippers

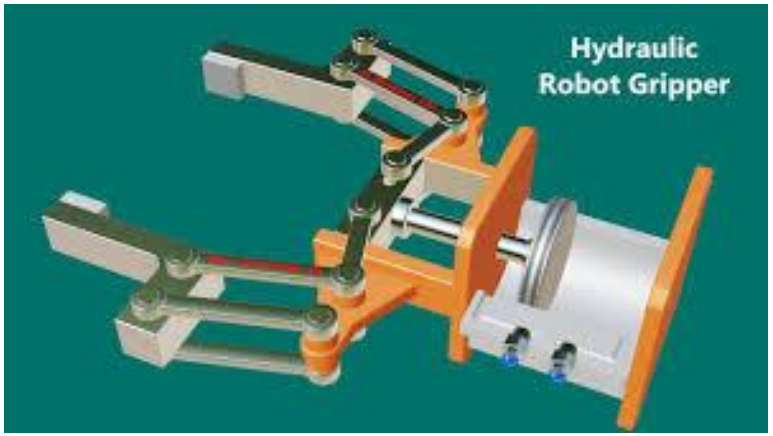
- **Principle**: Powered by compressed air, pneumatic grippers use air pressure to open and close their jaws. These grippers are fast, cost-effective, and simple to operate.
- **Applications**: Ideal for high-speed pick-and-place operations, packaging, and handling lightweight objects in automation and manufacturing.





Hydraulic Grippers

- **Principle**: Powered by hydraulic fluid, hydraulic grippers generate more force than pneumatic grippers, making them suitable for handling heavier objects.
- **Applications**: Suitable for applications requiring high gripping force, such as in heavy machinery or construction equipment where handling large, heavy parts is required.



Advantages and Disadvantages

- **Pneumatic Grippers**: Fast and lightweight but limited in force.

- **Hydraulic Grippers**: Offer high force but are slower, bulkier, and require more complex infrastructure.

Magnetic Grippers

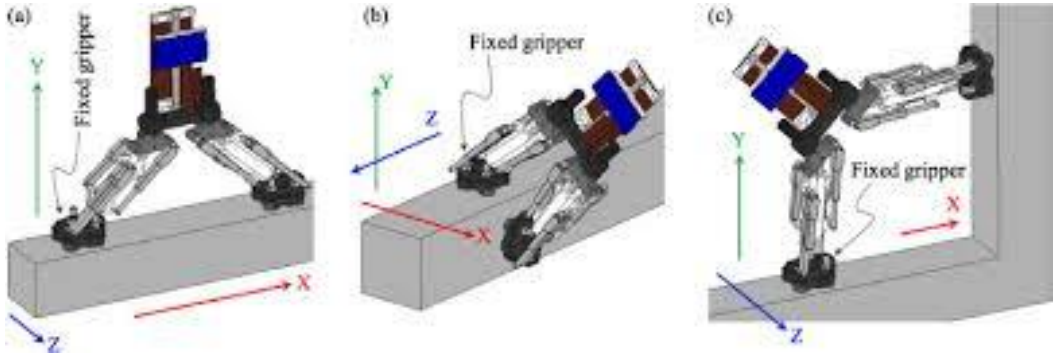
Principle

- Magnetic grippers use magnetic force to hold and manipulate ferromagnetic materials. The magnets can be **permanent** or **electromagnetic**, with the latter being controllable (able to switch on/off the magnetic force).

Applications

- Used in handling metal sheets, steel parts, and components in industries like automotive manufacturing and metalworking.
- Often employed in welding, metal cutting, and sheet metal processing.





Advantages and Disadvantages

- **Advantages**: Non-contact gripping method, ideal for handling thin, flat, or awkwardly shaped metallic objects.

- **Disadvantages**: Limited to ferrous materials and may require complex control systems for electromagnetic versions.

Vacuum Grippers

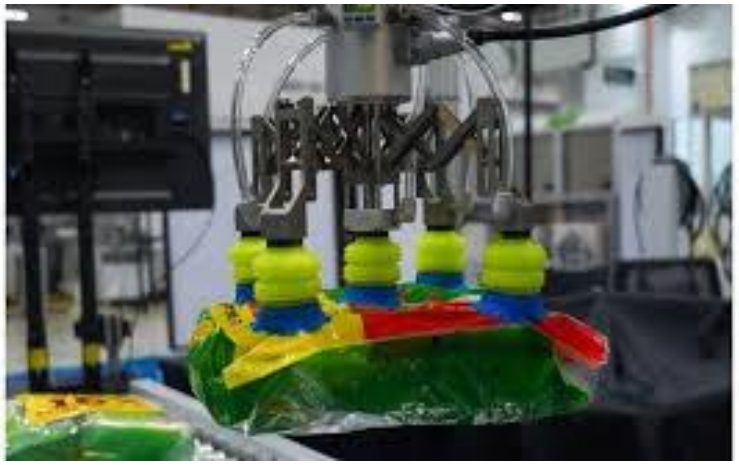
Principle

- Vacuum grippers use suction cups connected to a vacuum system to hold objects. They work by creating a pressure differential, where a lower pressure inside the suction cup compared to the ambient pressure causes the object to be gripped securely.

Applications

- Commonly used in packaging, logistics, and automation industries for picking up objects like glass panes, electronic components, and plastic materials.
- Effective in handling flat and smooth surfaces, such as sheets of material, paper, or glass.





Advantages and Disadvantages

- **Advantages**: Non-contact, can handle delicate or fragile items without damaging them.

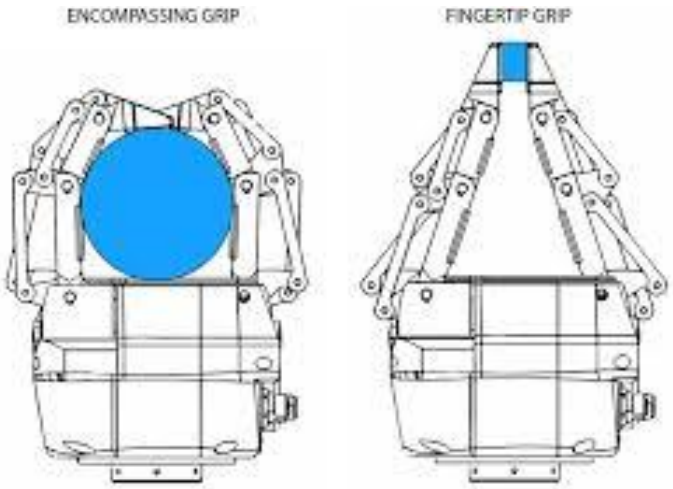
- **Disadvantages**: Limited to objects with smooth surfaces; requires constant air pressure and monitoring to maintain grip.

Advanced Grippers

1. Adaptive Grippers

Principle: These grippers can adapt to the shape and size of the object being grasped. Their fingers conform to the shape of the object, providing a secure grip on irregularly shaped items.

Applications: Used in robotics for handling objects of various shapes without the need for reconfiguration. Suitable for flexible manufacturing environments.





2. Soft Robotics Grippers

- **Principle**: Made from soft, flexible materials that can grip objects gently without damaging them. Often powered by pneumatic actuators that change shape to envelop the object.
- **Applications**: Ideal for handling delicate items, such as food, textiles, or fragile products in the food industry and e-commerce.

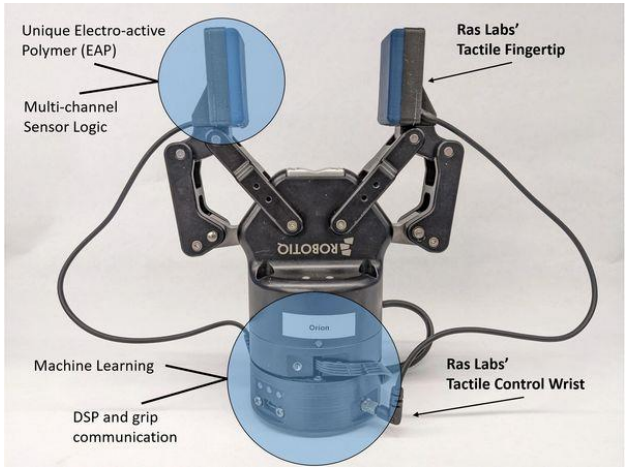


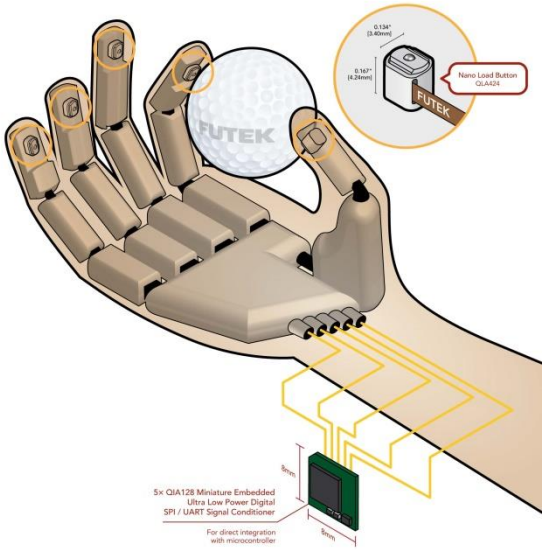


3. Tactile Sensor Grippers

Principle: Equipped with sensors that provide feedback on pressure, touch, and force distribution. This allows for more precise control over the gripping force, reducing the chance of damaging delicate objects.

Applications: Used in robotic surgery, precision manufacturing, and other fields where precise force control is crucial.

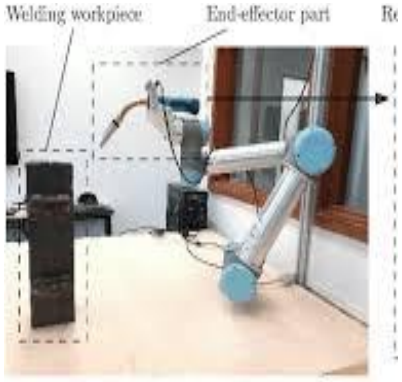




Process Tools as End Effectors

- **Welding tools**: For robotic welding.
- **Spray guns**: For robotic painting or coating.
- **Screwdrivers and drilling tools**: For assembly applications.
- **Cutting tools**: Used in material removal applications like milling or cutting.

These tools allow robots to perform specialized tasks beyond just gripping objects.



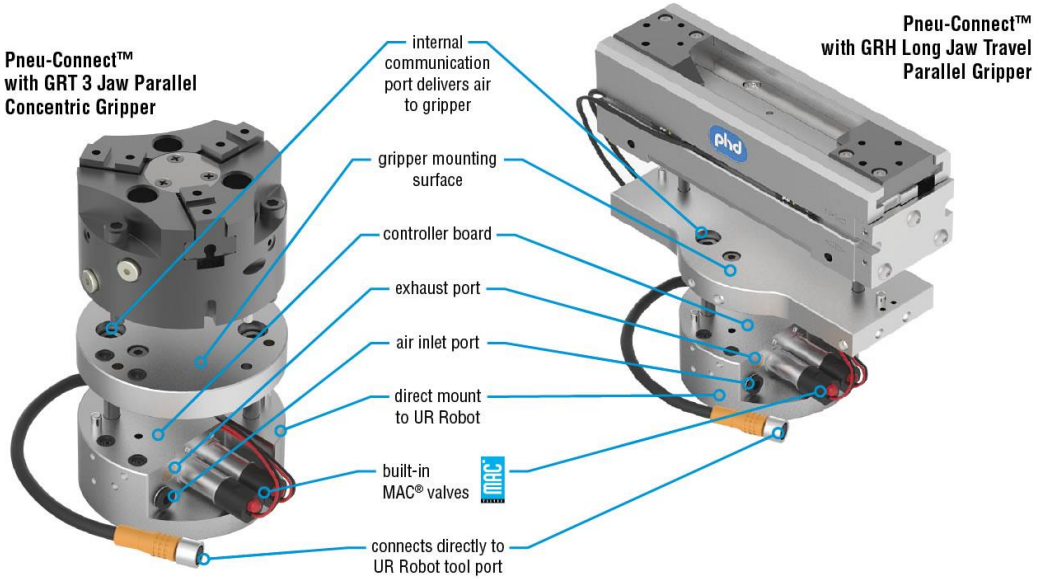






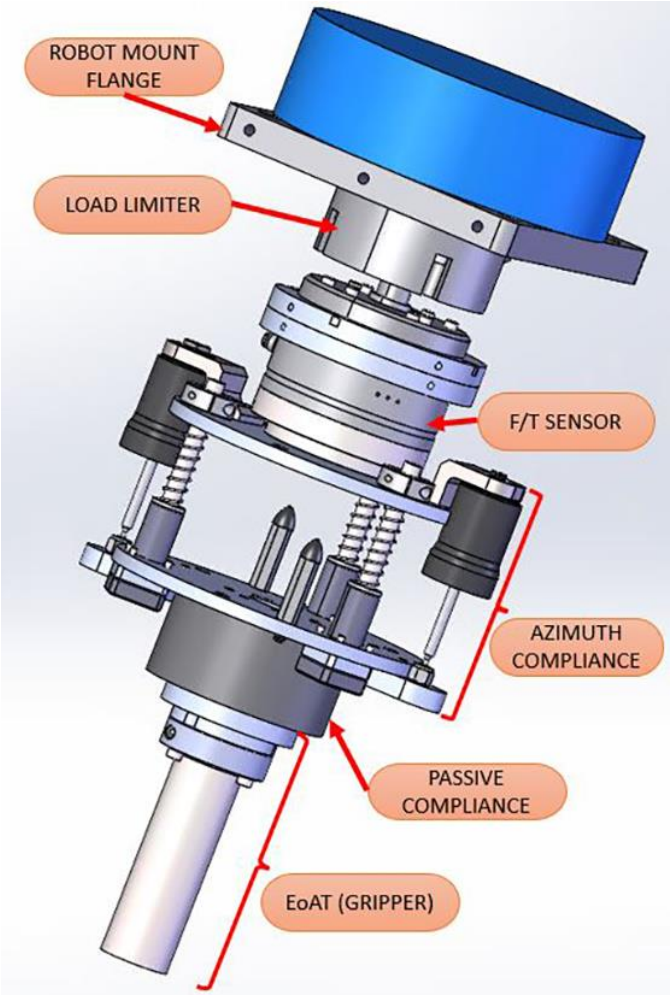
- **Robot End Effectors Interface**

The **end effector interface** is the connection point between the robot's arm and the end effector (gripper or tool). It includes mechanical, electrical, and communication interfaces that allow for control and quick swapping of different end effectors. Modern robots often use **tool changers** that allow for automatic switching between different tools and grippers.



Active and Passive Compliance

- **Active Compliance**: Refers to systems where sensors and actuators adjust the gripper's position or force in real-time based on feedback, allowing the robot to handle delicate or unpredictable objects with care.
- **Passive Compliance**: Uses mechanical or material properties (like springs) to allow slight movement or flexibility, enabling the gripper to adapt to variations in object shape or position without feedback.



Selection and Design Considerations

When selecting or designing a gripper, several factors need to be considered:

- **Object Characteristics**: Shape, size, weight, and material of the objects to be handled.
- **Type of Task**: Whether it's pick-and-place, manipulation, or delicate handling.
- **Force Requirements**: How much force is needed to securely grip the object without damaging it.
- **Environment**: Conditions like temperature, humidity, and cleanliness. For example, in food handling, the gripper must meet hygiene standards.
- **Speed and Precision**: How fast the gripper needs to operate and the level of precision required.
- **Robustness and Durability**: If the gripper will be used in harsh environments or high-wear conditions.